Module/Pathway Review: Bacterial Dimethylglycine and Sarcosine Catabolism in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00260 "Glycine, serine and threonine metabolism" **Module reviewed:** Two-step N-demethylation of dimethylglycine (DMG) → sarcosine → glycine **Module area:** amino_acid_metabolism

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1. Executive Summary

The two-step catabolic module — sequential N-demethylation of **dimethylglycine (DMG) → sarcosine → glycine** — is **fully satisfiable (COVERED)** in *Pseudomonas putida* KT2440. Both reactions are encoded within a single, synteny-conserved genomic cluster spanning **PP_0310–PP_0326**, which is directly orthologous to the experimentally characterized glycine-betaine catabolic system of *P. aeruginosa* (the *gbcAB*/*dgcAB*/*soxBDAG* loci). Step 1 (DMG → sarcosine) is performed by a **two-subunit, ETF-linked dimethylglycine dehydrogenase, DgcAB** (PP_0310/PP_0311) together with a dedicated electron-transfer flavoprotein EtfAB (PP_0312/PP_0313). Step 2 (sarcosine → glycine) is performed by a **tetrahydrofolate (THF)-coupled heterotetrameric sarcosine oxidase, SoxABDG** (PP_0323–PP_0326). A separate **monomeric sarcosine oxidase, PP_3775** (382 aa), provides an independent second route for step 2. Upstream, KT2440 supplies the module's ultimate substrate via choline → glycine betaine conversion by *betBA* (PP_5064/PP_5063).

The evidence for module satisfiability is **strong by synteny, orthology, and subunit-size transfer**, but it is important for curators to note that it is **not yet backed by KT2440-specific biochemical or mutant experiments**. The direct functional data come from *P. aeruginosa* (Wargo, Willsey) and from *Corynebacterium*/generic bacterial sarcosine oxidase biochemistry (Jorns). Transfer from *P. aeruginosa* is judged **strong** because of near-perfect gene order conservation and matching subunit architecture; transfer from *Corynebacterium* is **moderate** and used only to establish enzyme architecture (heterotetramer vs. monomer), not physiological role.

Two curation-critical caveats emerge. First, the **transcriptional regulators** that gate this module in *P. aeruginosa* — the AraC-family activators **GbdR** (induces *gbcAB*/*dgcAB*) and **SouR** (induces the *sox* operon) — have **no named ortholog** among the 66 candidate genes and are not annotated by name at either the *bet* locus or the catabolic cluster in KT2440; they are trans-encoded and currently unresolved. Second, the candidate list is dominated by **neighboring-pathway genes** of the broad KEGG ppu00260 overview map. Only ~14 of the 66 candidates have any plausible module role; three genes carrying a primary ppu00260 bucket — **PP_4421, PP_4423, PP_4432** — appear to be **over-propagated bucket assignments** with no DMG/sarcosine function. The single-chain dimethylglycine dehydrogenase alternative (DdhC of *Chromohalobacter salexigens*) is **not expected** in this taxon; KT2440 uses the two-subunit DgcAB architecture.

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2. Target-Organism Pathway Definition

2.1 Exact biochemical process included in the module

The module comprises exactly two oxidative N-demethylation reactions:

1. **DMG → sarcosine** (removal of one N-methyl group from dimethylglycine), catalyzed by an ETF-coupled dimethylglycine dehydrogenase (EC 1.5.8.-).
2. **Sarcosine (N-methylglycine) → glycine** (removal of the final N-methyl group), catalyzed by sarcosine oxidase (EC 1.5.3.1 for the classical oxidase; EC 1.5.3.24 for the 5,10-methylenetetrahydrofolate-forming heterotetramer).

Both reactions release a one-carbon unit. In the THF-coupled heterotetrameric sarcosine oxidase, the released C1 is captured as **5,10-methylene-tetrahydrofolate**; H_2O_2 is co-produced.

2.2 Neighboring pathways to keep separate

The following processes are **outside the module boundary** and should be kept distinct during curation, even though many of their genes appear in the ppu00260 bucket:

- **Upstream glycine-betaine demethylation** (glycine betaine → DMG), catalyzed by the Rieske-type glycine-betaine monooxygenase GbcAB (PP_0315/PP_0316). This is the *feeder* reaction, not part of the two-step module itself.
- **Choline → glycine betaine** (BetA/BetB; PP_5064/PP_5063) — an upstream supply route.
- **Downstream glycine disposal**: the glycine cleavage system (GcvP/H/T + Lpd) and serine hydroxymethyltransferase (GlyA), which consume glycine and the C1 units. These are explicitly outside the module.
- **Threonine biosynthesis** (ThrB, Hom, ThrC), **serine biosynthesis** (SerA/SerB/SerC), and various **serine/threonine dehydratases** (TdcG, IlvA paralogs) — these share the ppu00260 overview map but have no mechanistic connection to DMG/sarcosine catabolism.

2.3 Alternate names and database definitions

- KEGG map **ppu00260** = "Glycine, serine and threonine metabolism" — a **broad overview map**, not a module. The DMG/sarcosine steps are a small sub-portion.
- The *P. aeruginosa* orthologs are named after **glycine betaine catabolism** (*gbc*, *dgc*, *sox*, *sou*).
- The heterotetrameric enzyme is variously called **"tetrameric sarcosine oxidase" (TSOX)**, "sarcosine oxidase (5,10-methylenetetrahydrofolate-forming)" (EC 1.5.3.24), or **SoxBDAG/SoxABDG**.
- The single-domain enzyme is called **monomeric sarcosine oxidase (MSOX)** (EC 1.5.3.1).

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3. Expected Step Model

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    choline
      │  betA/betB (PP_5064 / PP_5063)      [upstream supply — outside module]
      ▼
  glycine betaine (GB)
      │  gbcAB (PP_0315 / PP_0316)          [feeder — outside module boundary]
      ▼
  dimethylglycine (DMG)
      │                                     ┌──────────────────────────────┐
      │  STEP 1: DMG → sarcosine            │  MODULE STEP 1               │
      │  dgcA/dgcB (PP_0310 / PP_0311)      │  ETF-linked DgcAB            │
      │  + EtfAB (PP_0312 / PP_0313)        │  EC 1.5.8.-                  │
      ▼                                     └──────────────────────────────┘
    sarcosine (N-methylglycine)
      │                                     ┌──────────────────────────────┐
      │  STEP 2: sarcosine → glycine        │  MODULE STEP 2               │
      │  ROUTE A: soxABDG                   │  THF-coupled heterotetramer  │
      │           (PP_0323–PP_0326)         │  EC 1.5.3.24                 │
      │  ROUTE B: PP_3775 (monomeric MSOX)  │  EC 1.5.3.1                  │
      ▼                                     └──────────────────────────────┘
    glycine  (+ 5,10-CH2-THF, H2O2)
      │  glyA (PP_0322), gcv system         [downstream sink — outside module]
      ▼
  one-carbon / central metabolism
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Both module steps have candidate genes; step 2 has **two independent architectures** encoded. The C1 by-products are disposed of by a conserved downstream signature: **PP_0327 (purU, formyltetrahydrofolate deformylase)** and **PP_0328 (formaldehyde dehydrogenase)**, immediately downstream of *soxG*.

4. Candidate Genes and Evidence

4.1 High-confidence module genes (the PP_0310–PP_0326 cluster)

Gene	Locus	UniProt	Role in module	Architecture evidence	Curation status
dgcA	PP_0310	Q88R24	DMG → sarcosine, catalytic subunit	686 aa flavoprotein	**Covered** (Step 1)
dgcB	PP_0311	Q88R23	DMG → sarcosine, catalytic subunit	650 aa flavoprotein	**Covered** (Step 1)
PP_0312	PP_0312	Q88R22	ETF α (electron acceptor)	410 aa	Covered (Step 1 partner)
PP_0313	PP_0313	Q88R21	ETF β (electron acceptor)	256 aa	Covered (Step 1 partner)
gbcA	PP_0315	Q88R19	GB → DMG (feeder, Rieske oxygenase)	EC 1.13.-.-	Outside module (upstream)
gbcB	PP_0316	Q88R18	GB → DMG (feeder, reductase)	EC 1.13.-.-	Outside module (upstream)
ltaE	PP_0321	Q88R13	L-threonine aldolase	EC 4.1.2.48	Boundary/co-located, not a demethylase
glyA1	PP_0322	Q88R12	SHMT (consumes 5,10-CH2-THF)	EC 2.1.2.1	Boundary (C1 coupling), outside module
soxB	PP_0323	Q88R11	Sarcosine oxidase β subunit	416 aa; EC 1.5.3.24	**Covered** (Step 2, Route A)
soxD	PP_0324	Q88R10	Sarcosine oxidase δ subunit	111 aa	**Covered** (Step 2, Route A)
soxA	PP_0325	Q88R09	Sarcosine oxidase α subunit	1004 aa; EC 1.5.3.1	**Covered** (Step 2, Route A)
soxG	PP_0326	Q88R08	Sarcosine oxidase γ subunit	210 aa	**Covered** (Step 2, Route A)
PP_3775	PP_3775	Q88GE9	Monomeric sarcosine oxidase	382 aa; EC 1.5.3.1	**Covered** (Step 2, Route B)

4.2 Step 1 — DMG → sarcosine (DgcAB + ETF)

Direct KT2440 proteome data (UP000000556) show that **DgcA (PP_0310, 686 aa)** and **DgcB (PP_0311, 650 aa)** are two large flavoprotein subunits, immediately followed by a dedicated **EtfA (PP_0312, 410 aa)** and **EtfB (PP_0313, 256 aa)**. This is the **two-component, "Pseudomonas-type" ETF-coupled dimethylglycine dehydrogenase** (EC 1.5.8.-), NOT a single polypeptide. The functional assignment transfers from *P. aeruginosa*, where mutations in *dgcAB* (PA5398–PA5399) abolish conversion of DMG to sarcosine [PMID: 17951379]. Because KT2440 encodes the two-subunit architecture with a dedicated ETF, the **single-chain dimethylglycine dehydrogenase alternative** (DdhC of *Chromohalobacter salexigens*) is **not expected** in this taxon.

4.3 Step 2 — sarcosine → glycine (SoxABDG heterotetramer)

The KT2440 gene order **glyA1(PP_0322)–soxB(PP_0323)–soxD(PP_0324)–soxA(PP_0325)–soxG(PP_0326)** reproduces the characterized *Corynebacterium* heterotetrameric sarcosine oxidase operon *glyA-soxBDAG* [PMID: 7543100]. This (αβγδ) enzyme carries both covalent and noncovalent FAD and **oxidatively demethylates sarcosine to glycine, H_2O_2, and 5,10-CH_2-tetrahydrofolate in an H_4folate/O_2-dependent reaction**. UniProt annotations are consistent: soxB (Q88R11) carries EC 1.5.3.24, "sarcosine oxidase (5,10-methylenetetrahydrofolate-forming)." Subunit sizes match the characterized TSOX profile: KT2440 SoxA (1004 aa) vs. TSOX α (967 aa); SoxB (416 aa) vs. TSOX β (405 aa) [PMID: 7543100]. Functional role transfers from *P. aeruginosa*, where *soxBDAG* is required for sarcosine catabolism [PMID: 17951379].

4.4 Step 2, Route B — monomeric sarcosine oxidase (PP_3775)

PP_3775 (Q88GE9, 382 aa) is annotated as a standalone monomeric sarcosine oxidase (EC 1.5.3.1). Its size matches the MSOX class ("approximately 388 residues") [PMID: 7543100], and it lies outside the *soxABDG* operon. KT2440 therefore encodes **both** sarcosine-oxidase architectures — the THF-coupled heterotetramer and a standalone monomeric oxidase — giving redundancy for module step 2.

4.5 Upstream supply (outside module but relevant)

BetB (PP_5063) and **BetA (PP_5064)** convert choline to glycine betaine, the ultimate substrate feeding the pathway; KT2440 *betBA* is functionally required for choline → glycine betaine transformation [PMID: 17116241]. **GbcAB (PP_0315/PP_0316)** then demethylates glycine betaine to DMG (feeder reaction).

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 Regulatory gap (module gating, trans-encoded)

The module is transcriptionally gated in *P. aeruginosa* by two AraC-family regulators: **GbdR** (PA5380), required for growth on GB/DMG and for induction of *gbcA/gbcB/dgcAB* [PMID: 17951379], and **SouR** (PA4184), the sarcosine-specific activator of the *sox* operon, required for appreciable growth on sarcosine as C/N source [PMID: 26503852]. A direct UniProt scan of KT2440 found **no named GbdR or SouR ortholog** among the 66 candidates. The *bet* neighborhood (PP_5058–PP_5068) and the catabolic cluster (PP_0308–PP_0328) contain **no AraC-family regulator** — intervening ORFs are chemotaxis MCPs (PP_0317, PP_0320) and hypotheticals. A genome-wide scan returned ~10 AraC-family regulators (e.g., PP_0298, PP_3665, PP_3659, PP_4508, PP_4511, PP_4852, PP_3149, PP_2425, PP_2173, PP_0876), none named *gbdR*/*souR*. **Regulation is trans-encoded and unresolved** — a real curation gap.

5.2 Likely over-propagated ppu00260 assignments

Three genes carry a primary bucket of kegg:ppu00260 but have no plausible DMG/sarcosine role and are not in the catabolic cluster:

Gene	Locus	Annotation	Assessment
PP_4421	PP_4421	Aminotransferase (EC 2.6.1.-, broad)	Over-propagated; broad EC, no module role
PP_4423	PP_4423	Succinylglutamate desuccinylase / aspartoacylase domain	Over-propagated; no module role
PP_4432	PP_4432	Xaa-Pro aminopeptidase	Over-propagated; no module role

These appear to be **over-inclusive bucket assignments** and should be **excluded from the module**.

5.3 Neighboring-pathway genes (majority of candidates)

Of the 66 candidates, the majority belong to separable neighboring processes of the broad ppu00260 map: threonine biosynthesis (*thrB* PP_0121, *hom* PP_1470, *thrC* PP_1471, PP_0662, PP_0664), serine biosynthesis (*serA* PP_5155, *serB* PP_4909, *serC* PP_1768), the glycine cleavage system (*gcvP1/H1/T-I* PP_0986–0989, *gcvP2/H2/T* PP_5192–5194, *lpd* PP_5366), serine/threonine dehydratases (*tdcG* PP_0297/0987/3144, *ilvA* PP_3446/5149, PP_3191/PP_4430/PP_2930), and phospholipid/one-carbon side reactions. **None of these are module enzymes.**

5.4 Paralog/architecture ambiguity to flag

- **glyA1 (PP_0322)** is physically embedded in the *sox* operon and functionally coupled to sarcosine oxidase (it consumes the 5,10-CH_2-THF product), but it is a **boundary/one-carbon enzyme**, not a demethylation step [PMID: 7543100]. **glyA2 (PP_0671)** is a second SHMT paralog elsewhere in the genome — do not conflate.
- **SoxB shares homology** with the N-terminal half of dimethylglycine dehydrogenase and with monomeric sarcosine oxidases [PMID: 7543100]; this homology is the root cause of potential annotation over-propagation between DgcAB, SoxB, and PP_3775. Curators should keep the **operon context** as the deciding feature.

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6. Module and GO-Curation Recommendations

6.1 Module step status

Module step	Status	Basis
Step 1: DMG → sarcosine	**covered**	DgcAB (PP_0310/PP_0311) + EtfAB (PP_0312/PP_0313); orthology + synteny + subunit sizes
Step 2: sarcosine → glycine (Route A, heterotetramer)	**covered**	SoxABDG (PP_0323–PP_0326); operon architecture + EC 1.5.3.24
Step 2: sarcosine → glycine (Route B, monomeric)	**covered (redundant)**	PP_3775 (382 aa, EC 1.5.3.1)
Single-chain DMG dehydrogenase alternative	**not_expected_in_target_taxon**	KT2440 uses two-subunit DgcAB, not DdhC-type
Module regulation (GbdR/SouR)	**candidate_uncertain / gap**	No named ortholog in candidate list; trans-encoded, unannotated

Overall module verdict: COVERED.

6.2 Module boundary recommendations

- The generic module boundary (DMG → sarcosine → glycine, excluding upstream GB demethylation and downstream glycine cleavage/serine conversion) is **correct for this organism** — no boundary revision needed for the core two steps.
- Recommend **explicitly documenting the two alternative architectures for step 2** (heterotetramer SoxABDG vs. monomeric MSOX) as both are realized in KT2440.
- Recommend **noting the trans-encoded regulatory dependency** (GbdR/SouR-type) as a module annotation, even though the regulators are outside the operon.

6.3 GO-curation

- Assign SoxABDG subunits to GO "sarcosine oxidase activity" and the 5,10-methylenetetrahydrofolate-forming reaction; DgcAB to dimethylglycine dehydrogenase activity (ETF-coupled).

- Flag PP_4421/PP_4423/PP_4432 for **removal of ppu00260 module association** (retain their genuine independent annotations).
- No new GO term request appears strictly necessary; existing EC 1.5.8.-, 1.5.3.1, and 1.5.3.24 terms cover the reactions.

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7. Genes to Promote to Full `fetch-gene` Review

Prioritized for full individual review:

1. **PP_0310 (dgcA)** and **PP_0311 (dgcB)** — confirm ETF-coupled DMG dehydrogenase catalytic assignment and EC 1.5.8.- refinement.
2. **PP_0323–PP_0326 (soxBDAG)** — confirm heterotetramer stoichiometry and the EC 1.5.3.24 (THF-forming) designation vs. generic 1.5.3.1.
3. **PP_3775** — confirm genuine monomeric sarcosine oxidase and rule out over-propagation from operon homology.
4. **Candidate regulator search** — promote a targeted review to identify the KT2440 GbdR/SouR functional equivalents among the ~10 AraC-family regulators (start with those genomically or regulon-linked; e.g., PP_0298 lies near the cluster).
5. **PP_4421, PP_4423, PP_4432** — promote for **de-association** review (confirm they should be dropped from the module bucket).

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8. Mechanistic Model and Interpretation

The KT2440 locus reconstructs, almost gene-for-gene, the *P. aeruginosa* glycine-betaine catabolic devolution: choline → glycine betaine (BetBA) → DMG (GbcAB) → sarcosine (DgcAB+ETF) → glycine (SoxABDG). The clustering of *ltaE* and *glyA1* within the *sox* region is not incidental — SHMT (GlyA1) directly recycles the 5,10-CH_2-THF released by the THF-coupled sarcosine oxidase, tying the demethylation module to one-carbon metabolism. The conserved downstream **purU (PP_0327)** and **formaldehyde dehydrogenase (PP_0328)** provide a disposal route for excess C1 units and formaldehyde, reproducing the syntenic *purU* signature reported downstream of the *Corynebacterium sox* operon [PMID: 7543100]. This C1-disposal signature strengthens confidence that the KT2440 cluster is a bona fide, coordinately organized demethylation pathway rather than a fortuitous gene neighborhood.

The redundancy at step 2 (heterotetramer + monomeric MSOX) suggests the organism can process sarcosine under multiple conditions, but only the operon-embedded SoxABDG is expected to be co-regulated with the upstream steps. The monomeric PP_3775 likely serves a distinct or overlapping physiological niche and may respond to a different (possibly SouR-like) signal.

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9. Evidence Base

PMID	Title (abbrev.)	How it supports the review
PMID: 17951379	Two gene clusters + regulator for *P. aeruginosa* glycine betaine catabolism	Direct functional proof that *dgcAB* converts DMG→sarcosine and *soxBDAG* is required for sarcosine catabolism; establishes GbdR as activator. Primary orthology source (transfer to KT2440 **strong**).
PMID: 7543100	Sequence analysis of sarcosine oxidase and nearby genes	Defines the *glyA-soxBDAG* heterotetramer operon, the THF/O_2-dependent reaction, subunit sizes, MSOX size class, and the downstream *purU* signature (transfer **moderate**, architecture only).
PMID: 26503852	Sarcosine catabolism regulated by SouR	Establishes SouR as the sarcosine-specific *sox* operon activator, required for growth on sarcosine — basis for the regulatory gap.
PMID: 17116241	Choline-O-sulphate utilization in *P. putida*	Direct KT2440 evidence that *betBA* produces glycine betaine — the upstream substrate feeding the module (transfer **direct**).
PMID: 19103776	GbdR regulates *plcH*/*pchP*	Corroborates GbdR as an AraC-family regulator responsive to choline catabolites, informing the regulator search.
PMID: 29703733	Glycine betaine monooxygenase (Rieske-type)	Characterizes the GbcAB-type oxygenase mechanism (upstream feeder).
PMID: 10689197	Stachydrine catabolism in *S. meliloti*	Comparative context for betaine N-demethylation via Rieske-type systems in other bacteria.

Strongest transfer: *P. aeruginosa* (near-identical gene order and architecture). **Direct KT2440 evidence:** limited to *betBA* function and to genomic/proteomic locus data. **No KT2440-specific mutant or enzymatic data** exist for *dgcAB* or *soxBDAG* themselves.

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10. Limitations and Knowledge Gaps

1. **No KT2440-specific functional experiments** for the core module enzymes (DgcAB, SoxABDG). All catalytic assignments rest on orthology/synteny to *P. aeruginosa* and architecture transfer from *Corynebacterium*. Confidence is high but not experimentally proven in the target strain.
2. **Regulators unresolved.** The GbdR/SouR functional equivalents in KT2440 are not identified by name; whether the module is inducible in KT2440 as in *P. aeruginosa* is inferred, not shown.
3. **Redundancy physiology unknown.** The relative contribution of SoxABDG vs. monomeric PP_3775 to sarcosine catabolism in KT2440 is undetermined.
4. **EC precision.** Whether the KT2440 heterotetramer is strictly the 5,10-CH_2-THF-forming enzyme (EC 1.5.3.24) versus the generic oxidase (EC 1.5.3.1) is inferred from soxB annotation and homology.
5. **Bucket noise.** The ppu00260 candidate list conflates the module with a large overview map; curators must apply operon/synteny filters manually.

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11. Proposed Follow-up Experiments and Curation Actions

1. **Growth phenotyping:** Test KT2440 growth on DMG and on sarcosine as sole C/N sources; construct *dgcAB* and *soxABDG* deletion mutants to confirm each step (mirrors the *P. aeruginosa* experiments).
2. **Regulator identification:** Use transcriptomics (RNA-seq on GB/DMG/sarcosine) plus promoter-reporter assays to identify which of the ~10 KT2440 AraC-family regulators activate the cluster; test PP_0298 first (proximity to the cluster).
3. **Enzymology:** Purify SoxABDG and confirm THF-dependence and 5,10-CH_2-THF production (distinguishing EC 1.5.3.24 from 1.5.3.1); assay PP_3775 for THF-independent sarcosine oxidase activity.

4. **Curation actions:** Mark module steps 1 and 2 as **covered**; mark the single-chain DMG dehydrogenase alternative as **not_expected_in_target_taxon**; document dual step-2 architectures; flag PP_4421/PP_4423/PP_4432 for de-association from the module; add a trans-encoded-regulator note.
5. **Promote to `fetch-gene` review:** PP_0310, PP_0311, PP_0323–PP_0326, PP_3775, and a regulator-candidate review.

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12. Key References

- Wargo MJ, et al. *Identification of two gene clusters and a transcriptional regulator required for Pseudomonas aeruginosa glycine betaine catabolism.* PMID: 17951379
- Chlumsky LJ, Zhang L, Jorns MS. *Sequence analysis of sarcosine oxidase and nearby genes reveals homologies with key enzymes of folate one-carbon metabolism.* PMID: 7543100
- Willsey GG, Wargo MJ. *Sarcosine Catabolism in Pseudomonas aeruginosa Is Transcriptionally Regulated by SouR.* PMID: 26503852
- Galvão TC, et al. *Uncoupling of choline-O-sulphate utilization from osmoprotection in Pseudomonas putida.* PMID: 17116241
- Wargo MJ, et al. *GbdR regulates Pseudomonas aeruginosa plcH and pchP transcription in response to choline catabolites.* PMID: 19103776
- *Glycine Betaine Monooxygenase, an Unusual Rieske-Type Oxygenase System.* PMID: 29703733
- *The stachydrine catabolism region in Sinorhizobium meliloti.* PMID: 10689197
